

More AI Reviewers Don't Help — Decorrelation Does

Large language models increasingly review production code. We built a data platform for the **Indigenomics Institute**, then used it to ask whether multi-agent AI reviewers actually beat a single reviewer — **validated on 30 real pull requests from the software we shipped for our industry partner**.

Companion to: “Evaluating Decorrelation in LLM Code Review: A Pre-Registered Study”

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THE RESEARCH CONTRIBUTION

Agreement predicts truth — but **only** when the reviewers are independent.

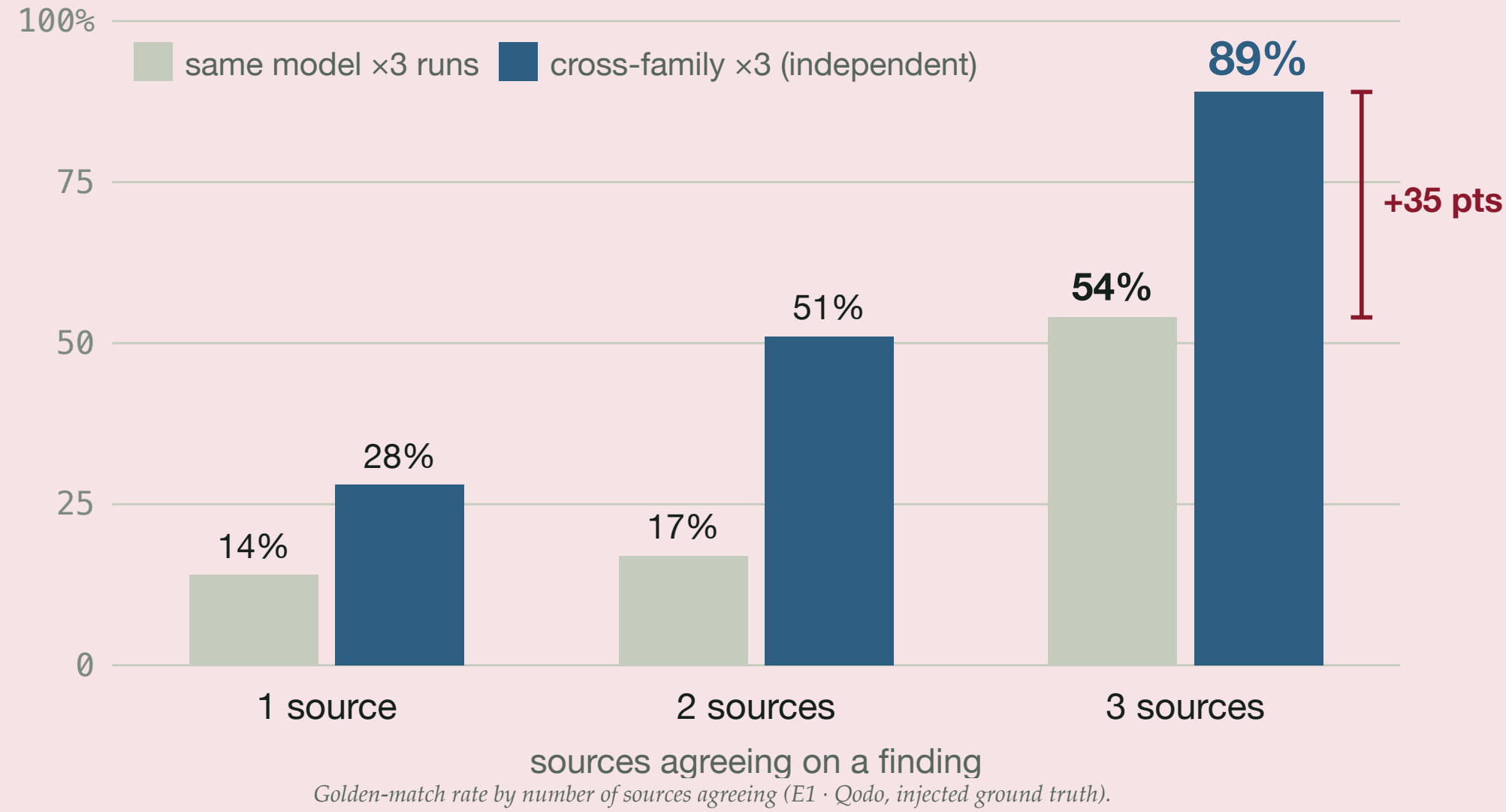
89%

three **independent** model families that agree match injected ground truth

54%

one model run three times — agreement is mostly noise

So what? If you are deploying AI code review, spend the budget on **independent verifiers** — different model families — not on more agents of the same kind. Agreement is only evidence when the reviewers could have failed differently.



01 THE CLIENT PROBLEM

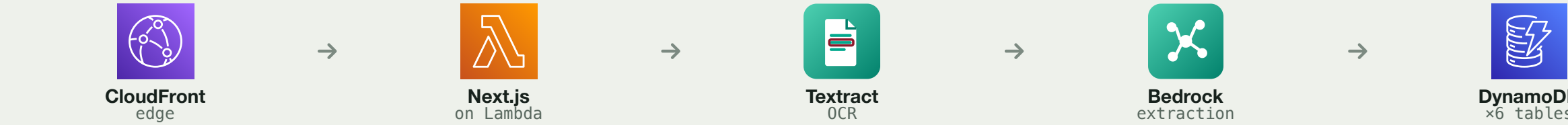
Canadian companies publish **Reconciliation Action Plan** commitments, and the legal precedents that shape them sit scattered across court records. There is no unified, searchable, verifiable view — so progress can't be tracked.

GROUNDING IN
Carol Anne Hilton's economic-reconciliation framework — an economic lens, not a case summarizer.

WHAT WE SHIPPED

AWS serverless deploy	2 data lenses	AI PDF extraction
106 commitments	100 organizations	561 legal decisions

HOW IT'S BUILT — SERVERLESS ON AWS



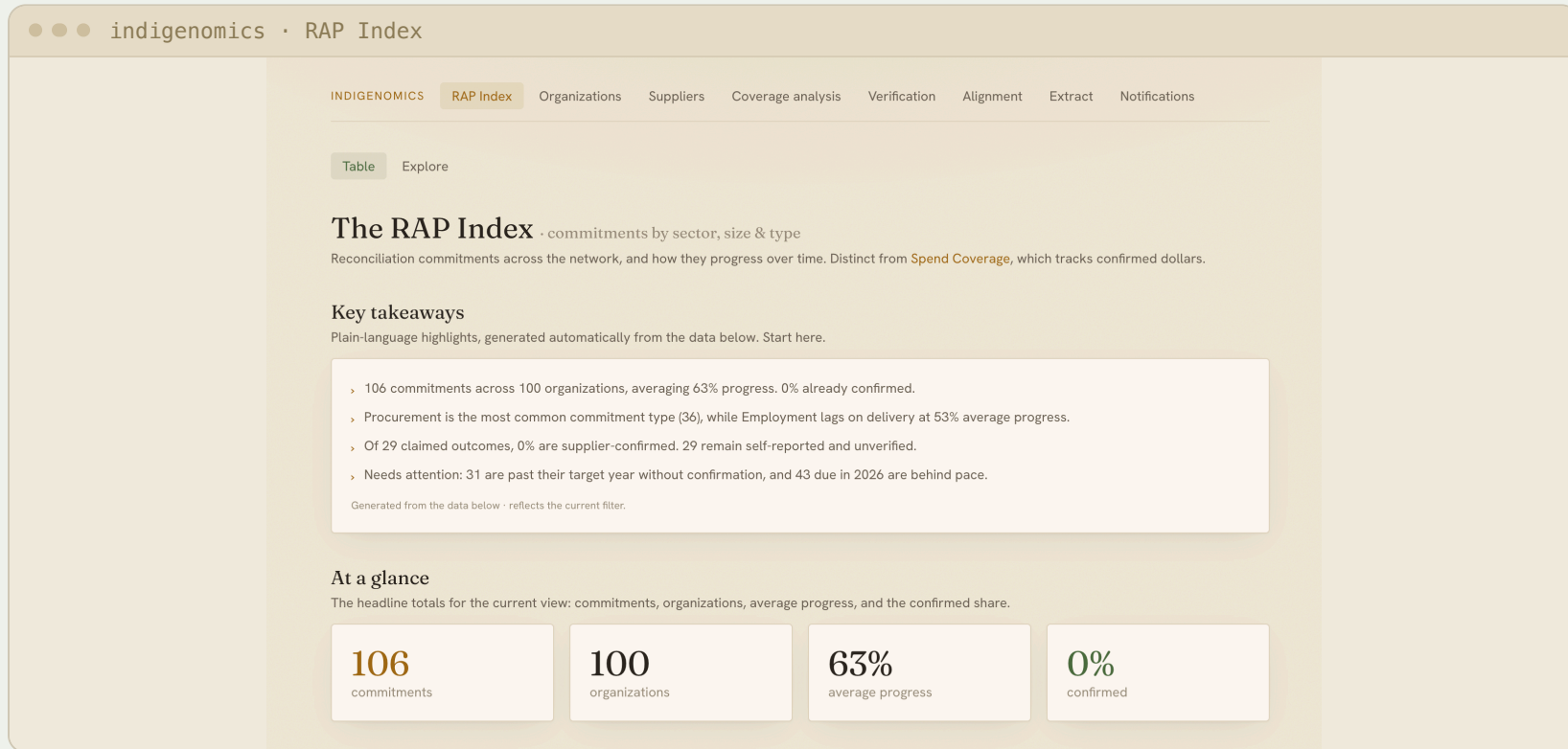
Data stays in *ca-central-1*; AI inference geo-routes to *us-east-1* — the OCAP residency boundary.

- WHAT THE INSTITUTE CAN NOW DO
- See every tracked commitment and its progress in one view
 - Get a weekly digest of what has gone **overdue**
 - Trace any claim back to the published source document
 - Read the legal precedent beside the corporate promise

02 THE RAP DATA PORTAL

WHAT IT DOES

- Searchable RAP commitments, scored by progress
- Indigenous legal-precedent database
- AI extraction from published PDFs
- Human review before anything publishes



RAP Index

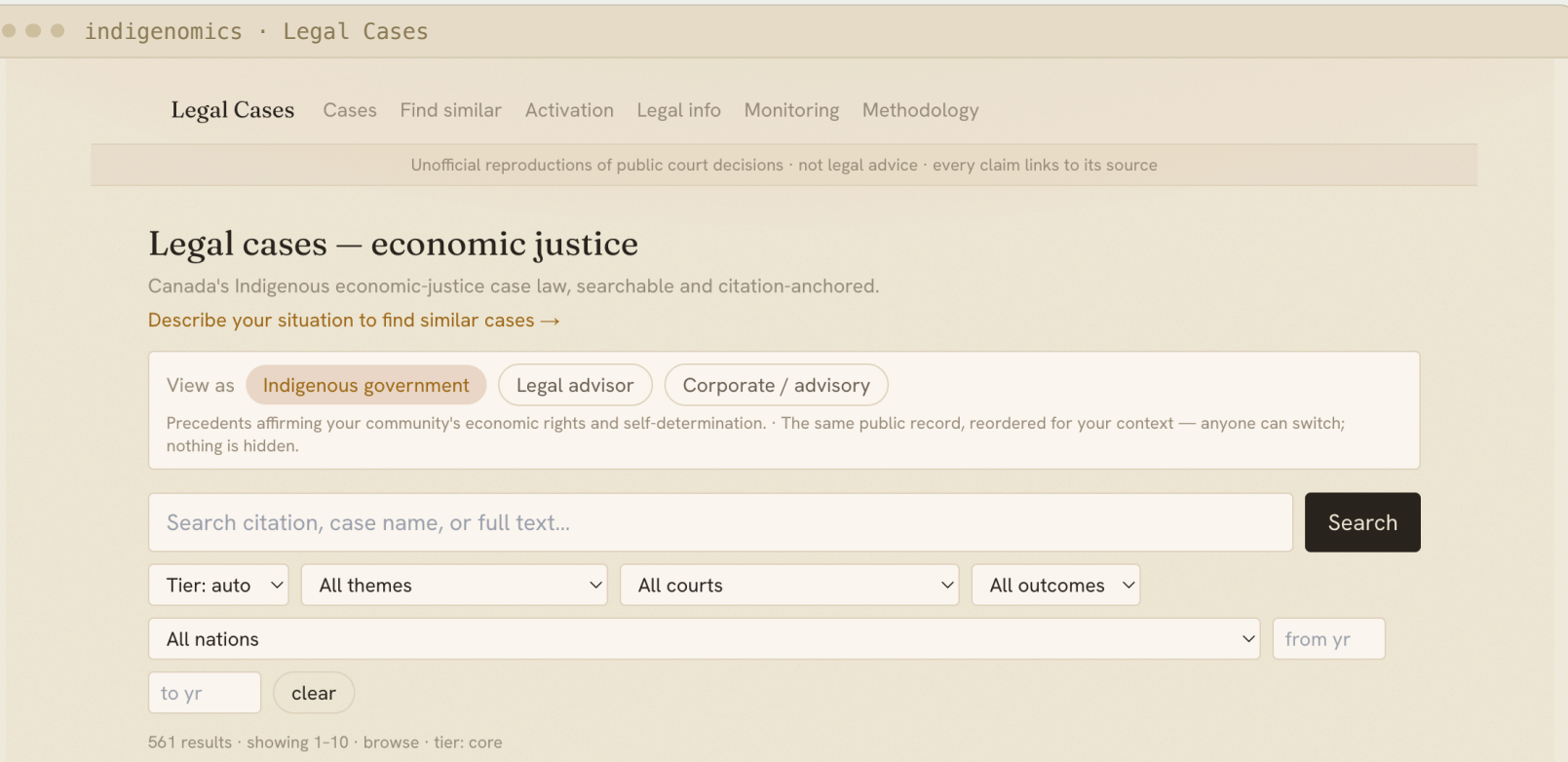
- 106 commitments, 100 organizations
- 63% average progress tracked
- auto-generated plain-language takeaways

HOW A COMMITMENT GETS IN



WHO USES IT

- Indigenomics Institute
- Indigenous organizations & nations
- Corporate partners reporting progress
- Researchers & legal advisors



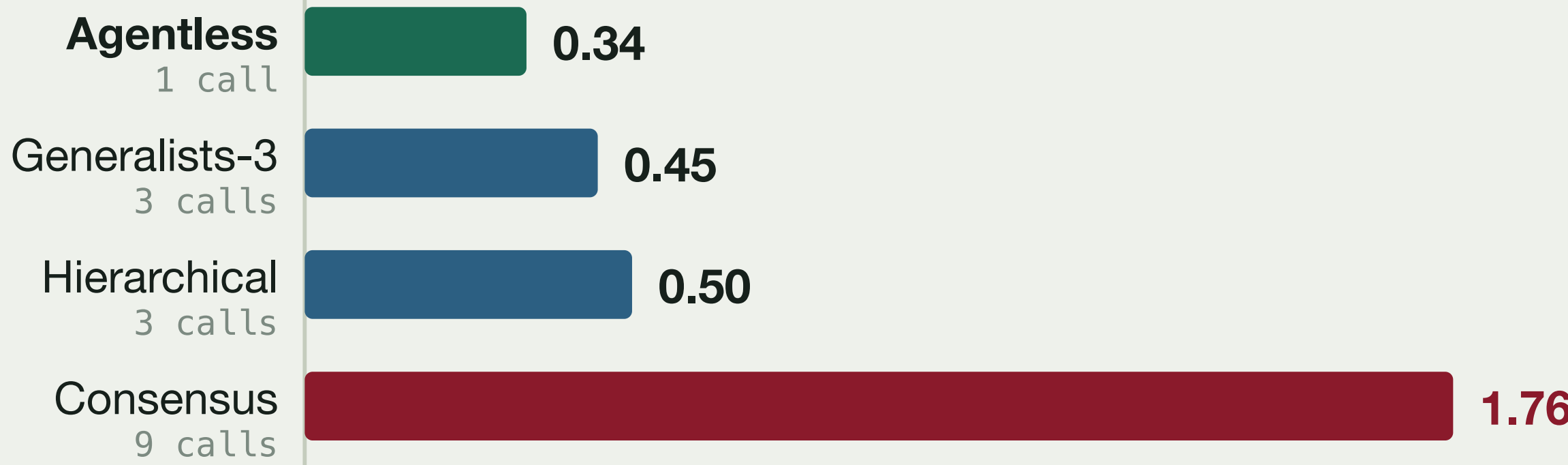
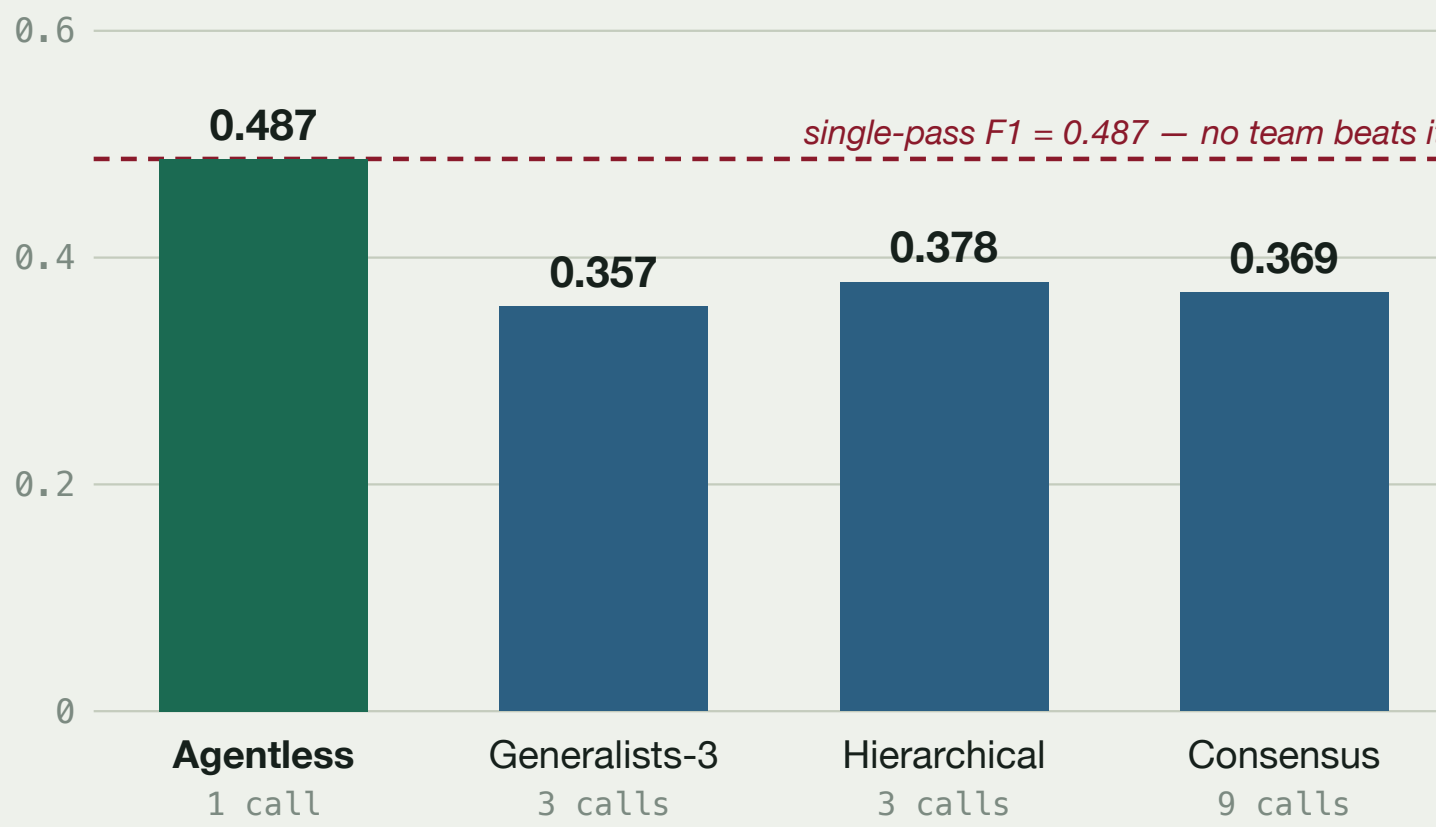
Legal Cases

- 561 economic-justice decisions
- citation-anchored, every claim sourced
- role-based views (nation · advisor · corporate)

03 RESEARCH FINDINGS · E1 · Q000, 99 PRS

Structure is null. No multi-agent arm beats single-pass F1; specialists tie compute-matched sampling ($p=0.73$); peer debate adds nothing over a manager ($p=0.12$).

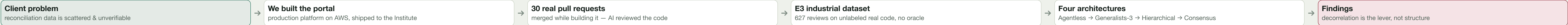
Decorrelation pays — on precision, not coverage (**89% vs 54%**). **Bounded:** recall plateaus at ≈ 0.83 ; on our own unlabeled code, LLM judges disagree ($\kappa=0.03$).



5x more expensive per confirmed finding than single-pass review

Extra agents buy recall at up to 5x the cost per confirmed findings.

04 HOW THE PROJECT BECAME THE RESEARCH



PROJECT IMPACT

A production platform delivered to the **Indigenomics Institute** — reconciliation commitments and legal precedents, verifiable and searchable in one place.

RESEARCH IMPACT

Don't deploy multi-agent review expecting a quality lift. **Decorrelate your verifiers** — independent model families — and read their agreement as a precision signal.

NEXT STEPS

Tool-grounded verifiers — execution and static analysis as a correctness oracle. **Open science:** all runs, ablations & artifacts released for replay.

